

HOMELAB

INFRASTRUCTURE OVERVIEW

azur3x.org
Proxmox VE 9.1 · HP Z420 Workstation
Updated: June 2026

HOST	HP Z420 Workstation
HYPERVISOR	Proxmox VE 9.1.6 · Kernel 6.17.13-2-pve
CPU	Intel Xeon E5-2697 v2 · 12C/24T · 2.70 GHz (boost 3.5 GHz)
RAM	64 GB ECC DDR3 · 8 × 8 GB · 1600 MT/s
STORAGE	ZFS mirror (tank) + ZFS single (backup) + LVM thin (sda)
DOMAIN	azur3x.org
NETWORK	Cloudflare Tunnels (services) + Tailscale mesh (remote access)

1. HARDWARE

Host Machine — HP Z420 Workstation

Component	Detail
CPU	Intel Xeon E5-2697 v2 @ 2.70 GHz · 12 cores / 24 threads · boost 3.5 GHz
RAM	64 GB ECC DDR3 · 8 × 8 GB · 1866 MT/s rated, running at 1600 MT/s
OS Drive	Samsung SSD 840 EVO 250 GB (sda) — Proxmox system + VM disks
NIC (primary)	Intel PCIe add-in card (nic1) — active bridge port for vmbr0
NIC (secondary)	Intel 82579LM onboard (nic0) — disabled (hardware hang under load)
Hypervisor	Proxmox VE 9.1.6 · Kernel 6.17.13-2-pve

Storage Disks

Device	Model	Size	Role	SMART / Notes
sda	Samsung 840 EVO	250 GB	System + VMs	✓ Clean · wear 4% · 99k POH
sdb	WD Red Pro WD4005FFBX	4 TB	tank mirror	✓ New · 0 POH · CMR · TLER
sdc	WD Red Pro WD4005FFBX	4 TB	tank mirror	✓ New · 0 POH · CMR · TLER
sdd	Hitachi HDT721010	1 TB	backup pool	■ 60k POH — plan replacement
sde	M3 Portable (USB)	1 TB	External HDD	Weekly backup archive · ext4

Previous disks retired June 2026: Seagate ST500DM002 (sdb, 66k POH, 16 reallocated sectors), WD WD10EZEX (sdc, 60k POH), Hitachi HDT721010 repurposed as single-disk backup pool. Tank rebuilt as 2× WD Red Pro 4TB mirror with ashift=12.

2. NETWORK

Topology

All traffic flows through a single Linux bridge (vbr0) on the host. VMs and LXC containers each receive a virtual ethernet pair connected to the bridge, sharing the physical uplink via nic1 (PCIe). nic0 (onboard Intel 82579LM) is disabled due to a hardware hang under VM bridge traffic.

Interface	Address	Role
vbr0	192.168.0.50/24	Main bridge — host + all guests share this uplink
nic1 (PCIe)	bridge port	Physical uplink — Intel PCIe NIC, active
nic0 (onboard)	—	Disabled — Intel 82579LM, hardware hang under load

Cloudflare Tunnels

Two cloudflared instances — one on the Proxmox host (local config), one inside CT 100 (dashboard-managed). No inbound firewall ports are open. All connections outbound over QUIC (UDP 443) to Cloudflare edge nodes.

Instance	Domain	Destination
Host (local config)	proxmox.azur3x.org	https://localhost:8006 — Proxmox web UI
CT 100 (dashboard)	nextcloud.azur3x.org	http://localhost:8080 — Nextcloud
CT 100 (dashboard)	jellyfin.azur3x.org	http://localhost:8096 — Jellyfin
CT 100 (dashboard)	tasks.azur3x.org	http://localhost:3456 — Vikunja
CT 100 (dashboard)	passwords.azur3x.org	http://localhost:8222 — Vaultwarden
CT 100 (dashboard)	homeassistant.azur3x.org	http://192.168.0.51:8123 — Home Assistant
CT 100 (dashboard)	grafana.azur3x.org	http://192.168.0.51:3000 — Grafana
CT 100 (dashboard)	photos.azur3x.org	http://192.168.0.55:80 — Hugo photo gallery

Tailscale Mesh

WireGuard-based overlay network for secure remote access. CT 106 acts as subnet router, advertising 192.168.0.0/24 to all mesh nodes.

Node	Role
CT 106 (tailscale-router)	Subnet router — advertises 192.168.0.0/24
Windows Desktop	Sunshine host — game streaming source (RTX 4070 Super)
Steam Deck OLED	Moonlight client — remote play target
Google Pixel 10 Pro XL	Mobile node
Dell XPS 15 (laptop)	University/mobile — HTTPS fallback on restricted networks

Guest IP Assignments

Guest	Type	IP	Notes
docker (100)	LXC	192.168.0.51	Static — main services host
torrents (101)	LXC	192.168.0.52	Static
dev-vm (102)	KVM	192.168.0.53	Static — virtio NIC on vbr0
rfid-wol (104)	LXC	DHCP	ACR122U USB passthrough
gallery (105)	LXC	192.168.0.55	Static — Hugo photo gallery
tailscale-router (106)	LXC	192.168.0.54	Static — Tailscale subnet router + Flask API

3. VIRTUAL MACHINES & CONTAINERS

VM 102 — dev-vm (KVM)

OS	Arch Linux
CPU	6 vCores (x86-64-v2-AES), 1 socket
RAM	8 GB
Disk	128 GB (LVM thin, ~10% used) · virtio-scsi
Network	virtio NIC on vmb0 · firewall=1 · IP 192.168.0.53
Access	SSH via dev-vm.azur3x.org (Cloudflare tunnel)
Purpose	Primary development environment — coding, university work

CT 100 — docker (LXC)

OS	Debian (privileged, nesting + fuse enabled)
CPU	8 cores
RAM	8 GB (peak usage ~1.8 GB)
Disk	40 GB rootfs (80% used) + ZFS bind mounts from /tank
GPU	/dev/dri passthrough — Jellyfin hardware transcoding
USB	ttyUSB0/1, ttyACM0/1 passthrough — Home Assistant hardware
Mounts	/tank/nextcloud, /tank/vaultwarden, /tank/vikunja, /tank/jellyfin, /mnt/backup
Startup	order=1, up=60

Container	Purpose	Port(s)
nextcloud + nextcloud-db	Self-hosted cloud storage (MariaDB)	8080
jellyfin	Media server (GPU hardware transcoding)	8096, 8920
vaultwarden	Bitwarden-compatible password manager	8222
vikunja + vikunja-db	Task management (PostgreSQL)	3456
homeassistant	Home automation	—
cloudflared	Cloudflare tunnel — all service domains	—
prometheus	Metrics collection	9090
node_exporter	Host-level metrics	9100
cadvisor	Container-level metrics	8080
grafana	Dashboards	3000
grafana-image-renderer	PNG panel rendering for reports	8081
alertmanager	Alert deduplication and routing	9093
alertmanager-discord	Discord webhook bridge	9094
diun	Docker image update notifications	—
searxng	Self-hosted meta search engine	8888

CT 101 — torrents (LXC)

OS	Ubuntu (unprivileged)
CPU	2 cores · RAM 512 MB · Disk 8 GB
Network	192.168.0.52 static

Purpose	Torrent client — downloads media to /tank/jellyfin for Jellyfin
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CT 104 — rfid-wol (LXC)

OS	Debian (unprivileged)
CPU	1 core · RAM 256 MB · Disk 4 GB
Network	DHCP
USB	ACR122U NFC reader — /dev/bus/usb/003/002 passthrough
Service	rfid-wol.py (Python3 + nfc-poll) — reads RFID tag → WoL to desktop
WoL target	Windows Desktop · MAC: bc:fc:e7:1c:87:bc

CT 105 — gallery (LXC)

OS	Debian (unprivileged)
CPU	2 cores · RAM 512 MB · Disk 8 GB
Network	192.168.0.55 static
Mount	/tank/gallery → /mnt/photos
Stack	Hugo static site — photos.azur3x.org
Design	Brutalist aesthetic — wordmark header, per-album colour tinting, ruled grid

CT 106 — tailscale-router (LXC)

OS	Debian (unprivileged)
CPU	1 core · RAM 512 MB · Disk 4 GB
Network	192.168.0.54 static · Tailscale mesh node
TUN	/dev/net/tun passthrough — required for WireGuard/Tailscale
Tailscale	Subnet router — advertises 192.168.0.0/24 to mesh
Flask API	HTTP control API :5000 — shutdown, reboot, WoL, Proxmox reboot
Client	Android HTTP Shortcuts — one-tap homelab control over Tailscale

4. STORAGE LAYOUT

ZFS Pools

Pool	Config	Size	Used	Notes
tank	2× WD Red Pro 4TB mirror (ashift=12)	3.62 TB	341 GB (9%)	Rebuilt June 2026 — new disks · 0 POH
backup	Single disk — Hitachi 1TB	928 GB	74.4 GB (8%)	■ 60k POH — plan replacement · no redundancy

ZFS Datasets

Dataset	Used	Mount / Purpose
tank/jellyfin	265 GB	/tank/jellyfin → CT101 /mnt/media
tank/nextcloud	74.3 GB	/tank/nextcloud → CT100 /mnt/nextcloud
tank/gallery	1.43 GB	/tank/gallery → CT105 /mnt/photos
tank/vaultwarden	10.5 MB	/tank/vaultwarden → CT100 /mnt/vaultwarden
tank/vikunja	96 KB	/tank/vikunja → CT100 /mnt/vikunja
tank/homeassistant	96 KB	/tank/homeassistant — configured, not yet mounted
backup/nextcloud	74.3 GB	Weekly ZFS send/receive from tank
backup/vaultwarden	10.5 MB	Weekly ZFS send/receive from tank
backup/vikunja	96 KB	Weekly ZFS send/receive from tank

LVM Thin Pool (sda → pve VG)

Volume	Allocated	Actual Use	Guest
pve/root	68 GB	~20 GB (31%)	Proxmox host OS
vm-100-disk-0	40 GB	~32 GB (80%)	docker CT — monitor usage
vm-101-disk-0	8 GB	~3.2 GB (40%)	torrents CT
vm-102-disk-0	128 GB	~12 GB (10%)	dev-vm
vm-104-disk-0	4 GB	~1.6 GB (40%)	rfid-wol CT
vm-105-disk-0	8 GB	~3.1 GB (39%)	gallery CT
vm-106-disk-0	4 GB	~1.6 GB (41%)	tailscale-router CT

Backup & Snapshot Schedule

Job	Schedule	Detail
ZFS daily snapshot	Daily 03:00	Recursive snapshot of full tank pool · 7 days retention
ZFS weekly backup	Sunday 02:00	Incremental ZFS send → backup pool · 14 tank snapshots · 2 backup snapshots kept
Docker config backup	Sunday 02:00	tar of CT100 /opt → /mnt/backup/docker-configs-DATE.tar.gz · 2 copies kept
ZFS TRIM	1st Sunday 00:24	tank pool
ZFS scrub	2nd Sunday 00:24	tank + backup pools
SMART monitoring	Continuous	smartctl_exporter → Prometheus → Grafana → Discord

5. SERVICES & STACK

Monitoring Stack

Component	Role
prometheus	Metrics collection — scrapes node_exporter, cadvisor
node_exporter	Host-level metrics — CPU, RAM, disk I/O, network
cadvisor	Container-level metrics — per-container resource usage
grafana	Dashboards — system overview, disk health, container stats
grafana-image-renderer	Renders panels as PNG for Discord reports
alertmanager	Alert deduplication and routing
alertmanager-discord	Discord webhook bridge — alerts and weekly summary
diun	Docker Image Update Notifier — alerts on new upstream images

Game Streaming Stack

Component	Detail
Host	Windows Desktop · RTX 4070 Super (NVENC) · Ryzen 7700x
Server	Sunshine — hardware-encoded game streaming
Client	Moonlight on Steam Deck OLED
Network	Tailscale mesh via CT 106 subnet router
Automation	Sunshine-AutoDisplaySwitch — handles display on stream connect/disconnect

RFID Wake-on-LAN

Component	Detail
Hardware	ACR122U NFC reader — USB passthrough to CT 104
Software	Python3 rfid-wol.py + nfc-poll daemon
Function	Reads RFID tag → sends Wake-on-LAN magic packet to desktop
WoL target	Desktop MAC: bc:fc:e7:1c:87:bc

Flask HTTP Control API (CT 106)

Accessible only over Tailscale. Android HTTP Shortcuts widgets for one-tap control.

Endpoint	Action
POST /shutdown	Shut down the Proxmox host
POST /reboot	Reboot the Proxmox host
POST /proxmox-reboot	Graceful Proxmox VE reboot
POST /wol	Wake-on-LAN magic packet → Windows Desktop

6. RESOURCE ALLOCATION SUMMARY

RAM

Guest	Allocated	Typical Use	Notes
docker CT (100)	8 GB	~1.8 GB	8 cores — GPU + USB passthrough
dev-vm (102)	8 GB	variable	6 vCores
torrents CT (101)	512 MB	< 200 MB	2 cores
rfid-wol CT (104)	256 MB	~13 MB	1 core — ACR122U passthrough
gallery CT (105)	512 MB	< 200 MB	2 cores — Hugo static site
tailscale-router (106)	512 MB	< 100 MB	1 core — Tailscale + Flask API
Total allocated	~17.8 GB	—	~46 GB free for host + ZFS ARC

ZFS ARC uses available free RAM as a read cache. With ~46 GB of host headroom, the ARC is well-fed for the tank pool workload (341 GB active data).

CPU

24 physical threads (12C/24T E5-2697 v2). vCPU allocation: 6 (dev-vm) + 8 (docker) + 2 (torrents) + 1 (rfid-wol) + 2 (gallery) + 1 (tailscale-router) = 20 vCPUs. CPU is shared dynamically — overcommit is normal and contention is unlikely in this workload profile.

LVM Thin Pool

Property	Value
Physical pool size	137 GB (sda3 → pve VG)
VG free	~16 GB
Total allocated	216 GB (thin overcommit — normal)
Actual used	~54 GB
Auto-extension	Configured at 80% threshold, expands 20% into remaining VG free

Note: vm-100-disk-0 at ~80% used — monitor and consider expanding CT 100 rootfs.